S2 Supplementary Data

Complete exploratory within-child age associations

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This supplement reports the artifact-masked re-export `neuralmanifolddynamics_figshare_infant_eeg_20260805_132612` (BAD_-interval artifact segments masked before epoching, 5,265 epochs; see S1.1). A prior, pre-masking export produced somewhat different numbers for several endpoints; the pre-masking values are noted inline where the change is informative (e.g. S2.5, S2.7) but are not reported in full here.

S2.1 Cohort-coordinate GEE family (primary result)

Results are stored in `results/figshare_infant_cohort_20260805_132612/03_cohort_coordinate_age/cohort_coordinate_gee_results.csv`. All 12 cohort-coordinate spline models were FDR-supported across 70 visits/22 children. The 3D coordinate results were mobility: $\chi^2 = 11.28$, FDR p=0.011; deviation: $\chi^2 = 54.72$, FDR p=$4.72 \times 10^{-}11$; and entropy-related coordinate: $\chi^2 = 15.09$, FDR p=$2.09 \times 10^{-}3$. The strongest 9D result was m_o ($\chi^2 = 155.37$, FDR p=$2.19 \times 10^{-}32$); the weakest overall was 9D d_s ($\chi^2 = 7.89$, raw/FDR p=0.048). They describe coordinate position in this frozen common chart and are the primary finding of this study (manuscript R2). This asymptotic GEE result should be read together with S2.2 (within/between decomposition) and S2.3 (cluster-permutation check), which qualify how robust each endpoint is.

S2.2 Within/between-child age decomposition (M3)

Results are stored in `cohort_coordinate_within_between_decomposition.csv`. At every one of the 12 endpoints, the within-child spline term (3 df) is significant: raw p ranges from $7.94 \times 10^{-}73$ (9D m_o, the strongest) to 0.0105 (9D d_s, the weakest). No endpoint's between-child linear term (1 df) is significant at $\alpha = 0.05$ after FDR correction; the smallest between-term raw p is 9D e_m at 0.041 (12-endpoint FDR ≈ 0.21), and most are far from significance (raw p up to 0.98 for 3D entropy-related coordinate). This directly addresses the potential between-child confound: the primary cohort-coordinate result is attributable to within-child movement, not simply to older and younger children differing cross-sectionally.

S2.3 Child-cluster row-permutation check (M4)

Results are stored in `cohort_coordinate_gee_cluster_permutation.csv` (500 permutations per endpoint, seed 42). Raw permutation p-values, smallest to largest: 9D m_o 0.002, 3D deviation 0.010, 9D d_n 0.016, 9D e_m 0.018, 9D m_e/e_e 0.020 each, 9D d_l 0.036, 9D m_a 0.052, 9D e_s 0.056, 3D entropy-related coordinate 0.130, 3D mobility 0.140, 9D d_s 0.222. After BH-FDR correction of these 12 permutation p-values, 6 of 12 endpoints remain supported: 9D m_o (FDR p=0.024); 3D deviation, 9D m_e, 9D d_n, 9D e_e, and 9D e_m (FDR p=0.040 each). Six endpoints do not survive: 9D d_l (FDR p=0.062), 9D m_a/e_s (FDR p=0.075 each), 3D entropy-related coordinate and 3D mobility (FDR p=0.152 each), and 9D d_s (FDR p=0.222). This is a materially more conservative picture than the asymptotic GEE (S2.1) and should be treated as the more trustworthy robustness statement given only 22 clusters.

S2.4 Cohort-coordinate direct-age metadata sensitivity

The spreadsheet-direct cohort-coordinate GEE subset included 40 visits from 22 children. Eleven of twelve endpoints remained FDR-supported; the 3D entropy-related coordinate was the only exception (raw/FDR p=0.070). In 3D, mobility and deviation had FDR p=$6.62 \times 10^{-}4$ and $3.10 \times 10^{-}9$, respectively. Note that 3D mobility is more strongly supported here than in the primary family, despite failing the cluster-permutation check (S2.3), whereas the 3D entropy-related coordinate now fails **both** this sensitivity check and the cluster-permutation check, making it the single least robust endpoint in the full family.

S2.5 Within-child MNJ and reachability family (robustness check)

All 70 sessions provided a resolved 9D MNJ; 20 children and 68 visits contributed after endpoint-complete within-child filtering. No primary 3D MNJ/reachability endpoint was FDR-supported; the smallest raw p was

for the MNJ trace (standardized slope −0.223; raw p=0.054; FDR sensitivity p=0.484), whose child-bootstrap interval excluded zero even though the sign-permutation p was just above the nominal threshold. This is not a uniformly clean null: the pre-artifact-masking export had a different smallest raw p (reachability effective dimension, 0.173), while the masked trace result is closer to nominal and should be interpreted with the permutation result and interval together. This is a distinct local-dynamics estimand from the cohort-coordinate GEE (S2.1) and is reported here as a robustness check, not as a primary result.

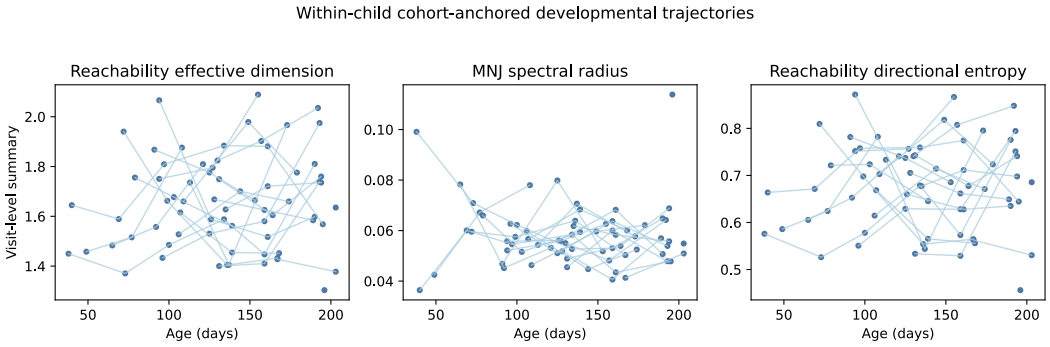


Figure 1: Within-child trajectories for selected primary summaries. Lines connect repeated visits but are not fitted developmental trajectories.

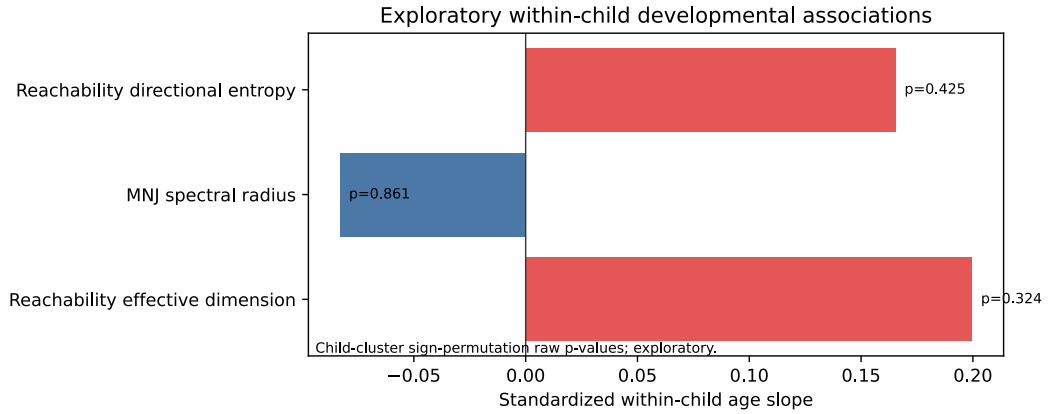


Figure 2: Exploratory within-child age effects. Labels show child-sign-permutation raw p-values; no primary endpoint was supported.

S2.6 Nine-dimensional MNJ sensitivity family

All 70 visits retained 9D MNJ summaries. This secondary 9D MNJ family (`within_child_age_associations_9d.csv`) is a distinct family from both the 9D cohort-coordinate GEE (S2.1) and the primary 3D within-child family (S2.5); it asks about local dynamics rather than chart position. Its own smallest raw p-value was for the Jacobian-condition-number endpoint (`mn9d_jacobian_cond_median`: standardized slope −0.32, raw p=0.076, FDR sensitivity p=0.212) – no longer nominally below 0.05 (the pre-masking export's smallest was `mn9d_aci_median` at raw p=0.015). No 9D MNJ sensitivity endpoint is FDR-supported.

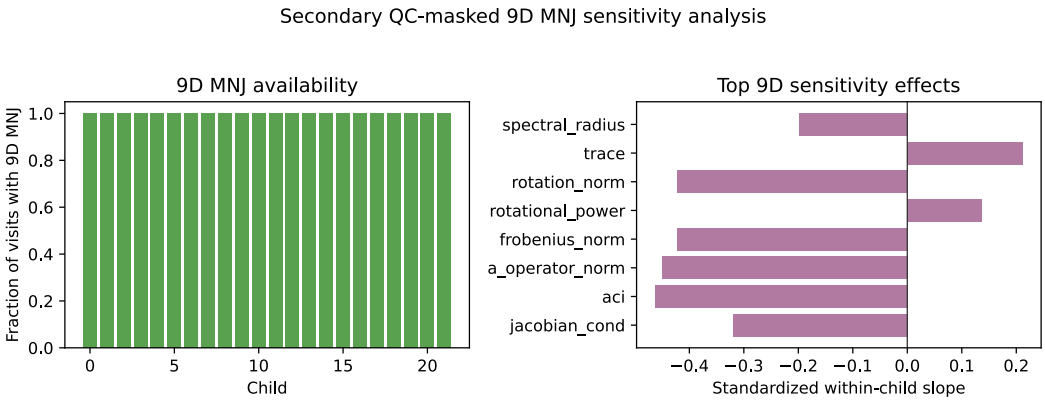


Figure 3: 9D MNJ availability and secondary within-child sensitivity effects. All visits had resolved 9D MNJs.

S2.7 Within-child direct-age metadata sensitivity

As a robustness check of S2.5, the primary 3D within-child family was re-run on the 36 visits from 18 children with spreadsheet-direct ages (`direct_age_metadata_sensitivity.csv`). No endpoint crossed nominal significance: the smallest raw p was `rotation_norm_median` at 0.208 (FDR sensitivity p=0.636). This is a materially different result from the pre-masking export, where the same two endpoints (`frobenius_norm_median`, raw p=0.050; `rotation_norm_median`, raw p=0.043) crossed nominal significance before FDR correction; both are now well above 0.2 (`frobenius_norm_median` raw p=0.277). Because artifact-flagged segments were specifically masked between the two exports, this pattern is consistent with the earlier nominal hits having been driven by residual motion artifact rather than a developmental effect, though this inference is descriptive (a before/after comparison across two non-independent exports of the same recordings), not a formal statistical test. It is reported here in full for completeness alongside the cohort-coordinate direct-age sensitivity (S2.4).

S2.8 Cohort-pooled reachability family (robustness check)

Results are stored in `results/figshare_infant_cohort_20260805_132612/05_cohort_pooled_reachability/`. All 70 visits were testable (`cohort_pooled_reachability_per_visit.csv`), drawing from a pooled set of 5,195 cross-session epoch-to-epoch transitions (down from 6,851 pre-masking) with a median of 8 distinct contributing children per visit after leave-one-child-out exclusion. None of the five endpoints was FDR-supported or nominally significant in the primary within-child-style test (`cohort_pooled_reachability_age_associations.csv`): rotational proxy (standardized slope −0.25, raw p=0.196, the smallest raw p in this family), directional entropy (raw p=0.277), effective dimension (raw p=0.351), log-volume (raw p=0.411), and anisotropy (raw p=0.478). This is a cleaner primary result than the pre-masking export, whose smallest raw p (anisotropy, 0.035) was nominally significant though not FDR-supported. The direct-age-only subset (36 visits/18 children, `cohort_pooled_reachability_direct_age_sensitivity.csv`) showed one nominal (uncorrected) hit – rotational proxy (standardized slope −0.47, raw p=0.040, FDR sensitivity p=0.200) – not FDR-supported. The per-visit neighbor k-th-distance was stable across age terciles (medians 0.352, 0.292, 0.329 in the youngest, middle, and oldest thirds), so these nulls are not attributable to a strong age-density confound in the pooled neighbor search. This is a distinct local-dynamics estimand from both the within-child family (S2.5-S2.7, estimated from each session's own epochs) and the cohort-coordinate GEE (S2.1, chart position rather than local flow).

S2.9 Result files

`first_to_last_change.csv` gives descriptive first-to-last within-child changes and age spans. It is not a shared-timepoint contrast because children entered and exited the study at different ages. Session-level source data and analysis manifests provide the complete auditable numeric record.

S2.10 Conventional-EEG (qEEG) comparator family: cohort-level GEE

Results are stored in `results/figshare_infant_cohort_20260805_132612/06_conventional_eeg_age/conventional_eeg_cohort_gee_results.csv` (S1.8; see that supplement for the second-export provenance note). Eighteen of 34 endpoints (all non-connectivity comparators) have 100% session coverage; the 16 connectivity endpoints have 80–96% coverage (S2.12). In the cohort-level spline GEE, 24 of 34 endpoints are FDR-supported: 14 of 18 non-connectivity endpoints (the four exceptions are the alpha/theta ratio, delta/alpha ratio, slowing index, and theta/alpha ratio) and 10 of 16 connectivity endpoints. Among the fully-covered endpoints,

the strongest are the beta/alpha ratio ($\chi^2 = 201.8$, FDR p=$1.94 \times 10^{-}42$) and relative beta power ($\chi^2 = 75.7$, FDR p=$1.49 \times 10^{-}15$). Among the connectivity endpoints, the alpha frontal-central coherence mean and SD ($\chi^2 = 297.1$ and 434.1, FDR p=$7.09 \times 10^{-}63$ and $3.05 \times 10^{-}92$) and theta frontal-central coherence mean and SD ($\chi^2 = 156.7$ and 93.4, FDR p=$7.91 \times 10^{-}33$ and $2.79 \times 10^{-}19$) are, by a wide margin, the largest cohort-level test statistics in this entire study; the alpha frontal-central pair is specifically flagged for a possible non-neural confound in the main text (Results R6, Discussion).

S2.11 Conventional-EEG within/between decomposition and cluster-permutation check

Results are stored in `conventional_eeg_within_between_decomposition.csv` and `conventional_eeg_cohort_gee_cluster_permutation.csv` (identical M3/M4 estimators as S2.2-S2.3, applied to this family, 500 permutations/endpoint, seed 42). The within-child spline term is FDR-supported at 31 of 34 endpoints; the between-child linear term is FDR-supported at only 1 of 34 (theta F8-P4 coherence SD, between raw p=$4.13 \times 10^{-}5$), so – as in the primary cohort-coordinate family (S2.2) – this family's cohort-level effect is overwhelmingly within-child rather than a between-child/cross-sectional artifact, including for the two most extreme frontal-central coherence endpoints (within raw p=$1.04 \times 10^{-}27$ and $8.18 \times 10^{-}7$; between raw p=0.22 and 0.16). The child-cluster row-permutation check is materially more conservative, as in S2.3: only 5 of 34 endpoints survive its own FDR correction – alpha frontal-central coherence mean and SD (permutation FDR p=0.0226 each), the beta/alpha ratio (0.0226), and relative beta and gamma power (0.0271 each). This is a comparably steep reduction from the asymptotic family (24/34 to 5/34) as seen for the primary cohort-coordinate family (12/12 to 6/12, S2.3), while endpoint count and the 500-permutation resolution also differ between families.

S2.12 Conventional-EEG within-child age associations and comparator coverage

Results are stored in `conventional_eeg_within_child_age_associations.csv` and `conventional_eeg_within_child_direct_age_sensitivity.csv` (identical S1.3 within-child estimator used for R4/R5, applied to this family; 68 endpoint-complete visits from 20 children primary, 36 visits from 18 children direct-age-only). Six of 34 endpoints are FDR-supported: permutation entropy (FDR sensitivity p=0.036), the beta/alpha ratio (0.007), the slowing index (0.044), relative beta power (0.034), relative gamma power (0.034), and theta frontal-central coherence mean (0.044). The spectral endpoint directions are consistent with prior reports of infant EEG spectral maturation (Marshall et al., 2002; Schaworonkow & Voytek, 2021; Xiao et al., 2018); complexity context is available from infant EEG work on multiscale entropy, including a preterm cohort that is not directly comparable to this sample (Pelc et al., 2022). The theta coherence result is reported without an expected-direction interpretation because it belongs to the lower-coverage connectivity pack and is not supported by the cluster-permutation check. The beta/alpha ratio and relative beta/gamma power are the only endpoints supported by **both** this check and the cluster-permutation check (S2.11), making them the most robust conventional-EEG endpoints overall. Comparator coverage (`conventional_eeg_coverage.csv`) is 100% (70/70 sessions) for all 18 non-connectivity endpoints and 95.7% (F4-P4 pair), 81.4% (F8-P4 pair), or 80.0% (central-parietal and frontal-central pairs) for the 16 connectivity endpoints; endpoints below the family's minimum-session/minimum-distinct-age/minimum-within-child-variation thresholds (S1.8) are marked `not_testable` rather than silently dropped or causing a fit failure, though no connectivity endpoint fell below those thresholds in this export.

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